Where does Qwen3-32B hold a paper's authors?

Stages 1–9

Small samples: find a signal, try to break it

20 → 50 → 200 papers (Stage 4, Pilot50, Stage 8 cohorts)

Stages 10–13 · post-defence additions (2026-11-05) at the end

Full cohort: locate it, then intervene

9,108 papers · 149,999 name tokens (Stage 10 cohort: papers.parquet, name_tokens.parquet)

One question, four lines of argument

Each line narrows the space of explanations; small samples set it up, the full cohort settles it

dark pages = Stages 1–9 · light pages = Stages 10–11 · purple tag = comparison

Where does Qwen3-32B hold a paper's authors, and can it say them?

A · This paper?

Paper-specific, not an easy name?

- 1–9 · Stage 5, 7.2, 7.3, 8
- 10–11 · title signal, name prior, probe

B · Readout tool

What can each readout see?

- 1–9 · Stage 3, 7.4, 8, 9
- 10–11 · lens comparison, linear probe

C · Which layer

When does it appear, when is it used?

- 1–9 · Stage 4, 7.2, 7.3, 8 (L56)
- 10–11 · fine layers, patching, knockout

D · Which token

Which part of name or prompt carries it?

- 1–9 · Stage 6, 7.1
- 10–11 · name format, tail tokens, co-authors

Stages 1–9 · 20–200 papers

Pilot design

- Pilot50 / Stage 8: 50 and 200 papers (pilot50_v1.csv, stage8_papers.csv)
- Mostly first-listed author, first token
- Score: gold vs 1 or 100 false names (logit margin, win rate)
- A/B both R0, matched in pairs
- Wrong titles: 1 crossed or 20 shuffled
- Layer L56 chosen after repeated looks

Stages 10–11 · 9,108 papers

Full-cohort design

- All 9,108 papers (papers.parquet), every author, every token
- Score: rank over the whole vocabulary, log10(1/rank)
- target = A + C, control = B + D (R0 + R1), compared within citation bins
- 5 matched false titles + no title
- All layers reported, fine layers added
- Parity and Stage 6 cross-check before analysis

What changed in the design, and why

Small samples could not separate "small effect" from "no effect" in a given layer. The full cohort trades candidate-based scores for full-vocabulary ranks and pays for it with GPU time. **Note:** target/control include R1 papers; with R0 only (A vs B, as in the pilot) the output gap is +0.199 vs +0.207 (Stage 10, final L63, R0 3,423 vs 2,151; §6.13) and every layer looks the same.

Same papers, two ways of grouping

Stages 1–9 compare A vs B; Stages 10–11 compare target vs control. Counts from the 6,841-paper cohort.(Stage 1, 351_audit.json group_counts)

	recognised (target)	not recognised (control)
R0 · no true author written	A = 4,110	B = 2,186
R1 · some authors written (HR2 > 0.5)	C = 511	D = 34
Total	**target = A + C = 4,621**	**control = B + D = 2,220**

Stages 1–9

A vs B: both R0, so the contrast is purely "knows but cannot say"

Stages 10–11

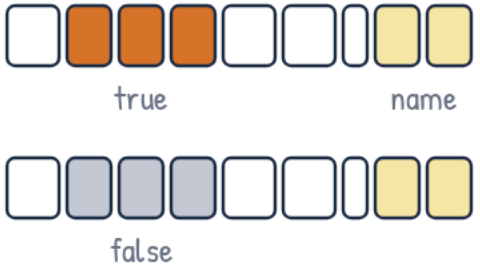
(A + C) vs (B + D): adds 511(351_audit.json) partial-recall papers, almost all on the target side

Check done: with R0 only (A vs B) the output gap is +0.199, versus +0.207 for target vs control(Stage 10, final L63, within-bin; §6.13), and every layer looks the same. The two phases compare like with like.

Reading the method figures

1 2 3 steps, in order

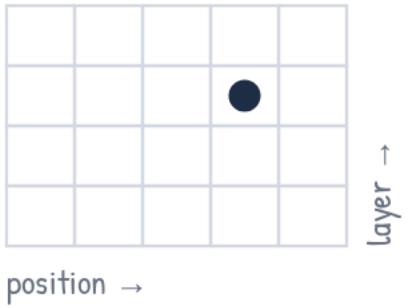
Prompt tokens



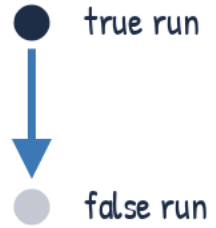
Layers



State = dot



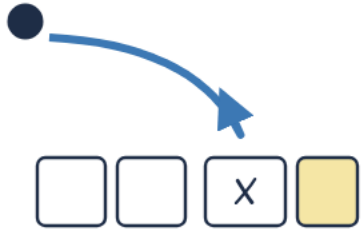
Copy a state



Block



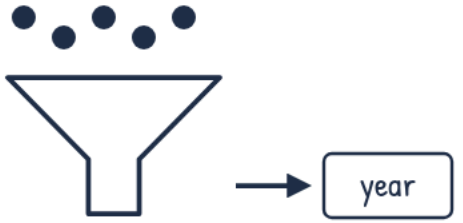
Inject



Read out



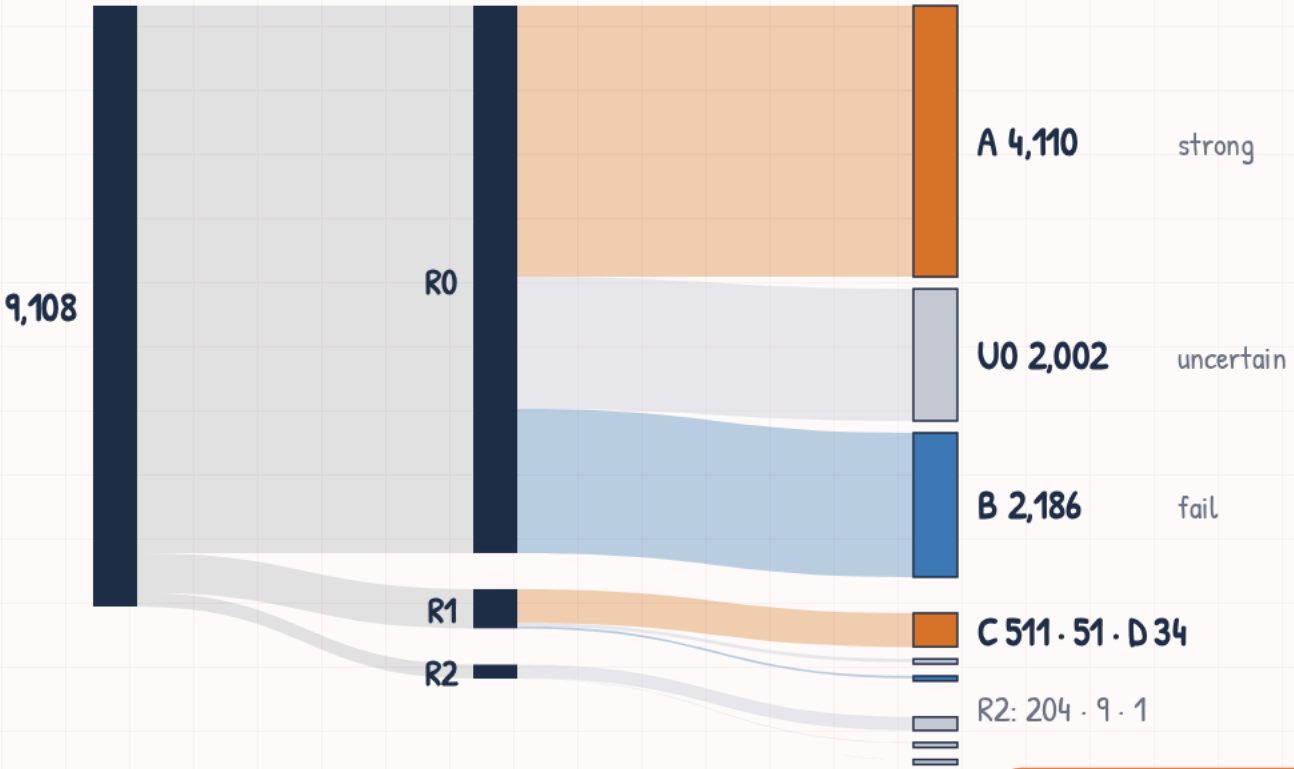
Probe



Groups and samples

example: cell C

1 papers → 2 open recall → 3 recognition → 4 cells → 5 samples



Stage 10 · M2 · M3

all papers

Stages 11–13 · X

593 target + 593 control

1,178 matched pairs

Stages 1–9

Pilot50 · Stage 8 (A, B)

target = A + C = 4,621 · control = B + D = 2,220

Counts: 351_stage10/cohort/papers.parquet (recall_level × MC outcome); 593 + 593: 361_stage11 sample (1,186 papers); 1,178 pairs: Stage 13 X1/X3 pair table

Cohort and behavioural labels

From the 15 Oct label package (9,108 papers) to a frozen 6,841-paper analysis cohort (351_audit.json total, four_group_total)

Group	Recognition / open recall	Full (351_audit.json)	Pilot50 (pilot50_v1.csv)
A	pass / R0 (no true author written)	4,110	20
B	fail / R0	2,186	20
C	pass / R1 (some authors matched)	511	7
D	fail / R1	34	3

- Pass = 4/4 option orders correct *and* mean true-option probability ≥ 0.5; fail = 0–1/4
- Pilot50: seed 35150, keeps the 20 Stage 4 papers + 30 new
- 20 A/B pairs matched on exact field and citation bin, then $|\Delta\text{year}| + 2 \cdot |\Delta\text{authors}|$

Why A vs B?

Both groups write **no** correct author. They differ only in whether the model recognises the true list. That isolates "knows but cannot say".

Caveat

In the 0–2-citation bin the strict pass rate is **25.5%** (796 papers; label_README.md l.8), above the planned ≤10% floor: "pass" is not proof of stored knowledge.

One fixed model, prompt and score

Everything later is measured under these settings

Model	Qwen/Qwen3-32B, BF16, revision 9216db57, 64 blocks, d = 5,120
Layers	L16, 24, 32, 40, 48, 56, 62 → target L63
Readouts	Public J-Lens · Vanilla (logit lens) · real final logits
Position	last input position; context limit 512
Names	encode(prompt) + encode(" " + author)
Hardware	29 blocks on GPU, 35 offloaded to CPU

> Who are the authors of the paper titled "{title}" ({year})?
> Authors:

Rank r = 1 + #tokens with a strictly larger logit

JMRR = mean over authors of mean over tokens of 1/r

Integrated = average of the 7 source layers (L63 excluded)

Guard: forced token IDs are checked, and a J-Lens/Vanilla final-logit gap > 0.1 rejects the run.

How groups and contrasts are built

labels: group lead's pipeline on the same Qwen3-32B · we use them unchanged

open answer -> R0 / R1 / R2

greedy, chat template

fuzzy name match: M, H, U

HR2 = (H+U) / (M+H+U)

R2: HR2 <= 0.5 · R0: M = 0 · R1: M > 0

a fixed rule, no human or model judge

multiple choice -> target / control

true list + 3 real teams of other papers

(same field, team size, citation bin, ~year)

4 orders · p(letter) from next token

target: 4/4 and mean p(true) >= 0.5

control: <= 1/4

5 false titles per paper

same field, own year kept

min-cost on |dyear|/5 + |dauthors|

+ |dtitle tokens| + 1.5 |dcitation bin|

no shared surname, Jaccard <= 0.6

79% strict tier (Stage 10, 35,898 of 45,540 pairs, conditions_manifest.json)

statistics

target vs control only within citation bin

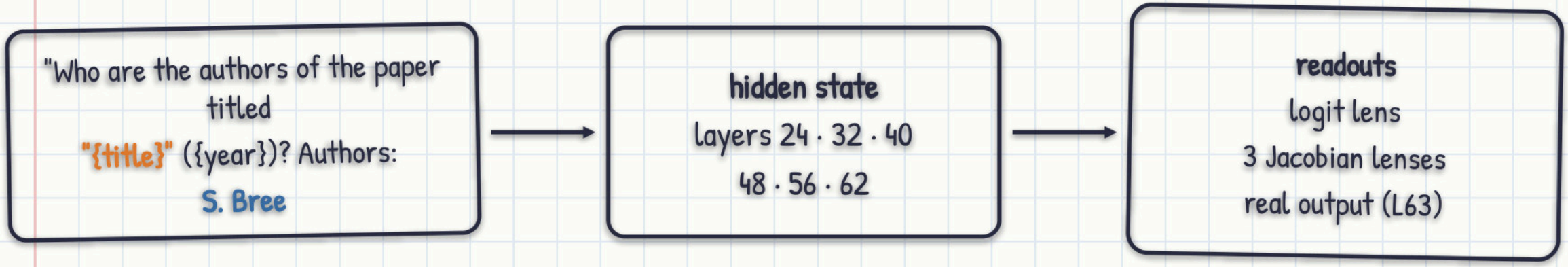
10 bins with >= 30 per group (Stage 10, §6.2)

weight $n_t n_c / (n_t + n_c)$

95% bootstrap over papers, 2,000

sensitivity: drop bins 0-4

How we read the model



Score = rank of the correct next name token,

as log10(1 / rank): 0 = rank 1, -1 = rank 10

7 title conditions: the true title, 5 matched false titles, no title

title-specific signal

= true title - mean(5 false titles)

Same name, same tokens: what is left is knowing

this paper

Lenses (model weights never change; Jacobians to L63 averaged over prompts): public = 557 WikiText-103 prompts (Neuronpedia) · self100 = 100 frozen WikiText-103 prompts · fitA = 100 other WikiText-103 prompts (Stage 9: one skipped prompt replaced -> 100 valid) (n_prompts: 351_stage10/config/lenses.json; Stage 9 final_status.json)

Can we trust the numbers?

fast = slow

One forward pass vs one per prefix: max gap 0.37 logit (bf16 rounding); ranks 1-41 identical (Stage 10, 12 contexts, parity.json)

matches Stage 6

972 tokens: same token ids, median log-prob gap ~1e-5 (Stage 10, crosscheck_stage6.csv)

negative control

Punctuation tokens show ~0 title-specific signal at every layer (Stage 10, max 0.014, 9,108 papers, §6.5)

Caveat: the parity tolerance was raised 0.25 -> 0.5 after seeing these bf16 gaps. (parity.json tolerance_amendment)

Line A

Is it about this paper?



small samples asked

Does the true title beat a crossed or shuffled title? Does A gain more than B?

→

the full cohort answered

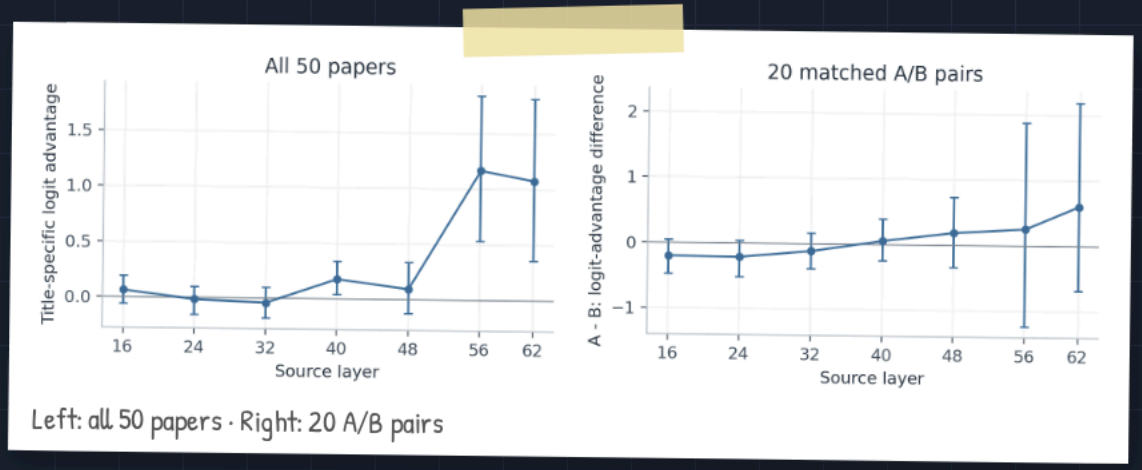
Title-specific signal for every token; target beats control only late; name prior split out; probe controls

Crossed title–author control

Is the gold name high because of the title, or because the name is easy?

Paper P (author G) and source paper Q (author F, same field & bin).
M = logit(G) − logit(F) · T = M(P title) − M(Q title)
A fixed preference for G cancels out.

- All 50 papers: L56 +1.17 [0.53, 1.84], L62 +1.08 [0.36, 1.82] (Stage 5, J-Lens, 50 papers, layer_summary.csv)
- Matched A–B, 7-layer integrated: +0.08 [−0.43, +0.65] (20 pairs, matched_ab_summary.csv)

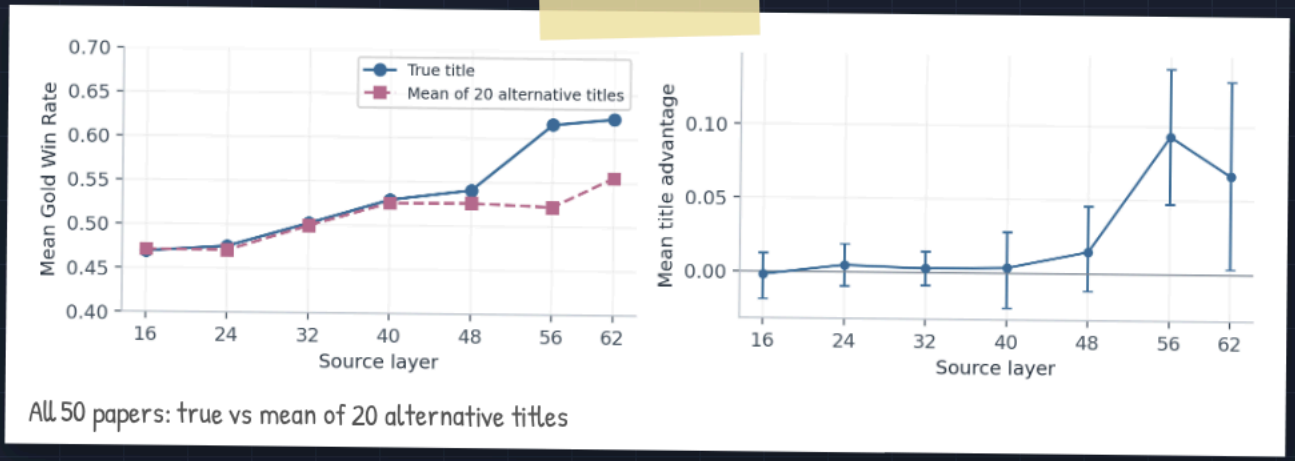


Left: all 50 papers · Right: 20 A/B pairs

The title changes which name wins in late layers. Whether A gains more than B: **not established**.

Shuffled titles: the true title helps

Same 100 candidates, 20 metadata-matched wrong titles per paper, 1,000 contexts; only the title changes



All 50 papers: true vs mean of 20 alternative titles

- Pooled L56: 0.616 vs 0.523 → +9.3 pp [4.8, 14.0], 35/50 papers positive (J-Lens L56, 50 papers; paper_null_distribution.csv; CI recomputed 10-30)
- Leave-one-out: +8.6 to +10.0 pp (stage7_l56_loo.csv)
- Vanilla L56 +10.6 pp, final output +7.2 pp (paper_null_distribution.csv)
- Matched A–B integrated: +0.018 [−0.021, +0.063] (20 pairs, integrated_ab_bootstrap.csv)

Two different questions: "does the title help?" **yes**; "does it help A more than B?" **not shown**.

Does the title effect survive the obvious objections?

Source-author overlap

199/1,000 wrong titles come from a paper whose author is in the false pool. Excluding them: +0.026 → +0.028.(quality.json; paper_null_distribution.csv)

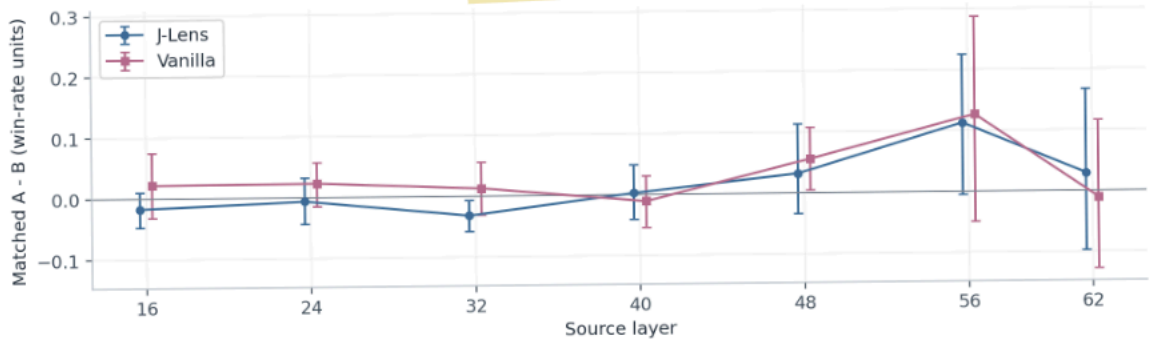
Candidate weighting

Author-, unique-token- and non-collision weights: pooled +0.026 to +0.032; all six A–B CIs include 0.

(stage7_weighting_title_advantage.csv)

Low citations

0–2 citations (n = 6): L56 +0.22, 5/6 positive, but 0/6 beyond its own 20-title tail (≤ 0.05).(floor_L56_low_six.csv)



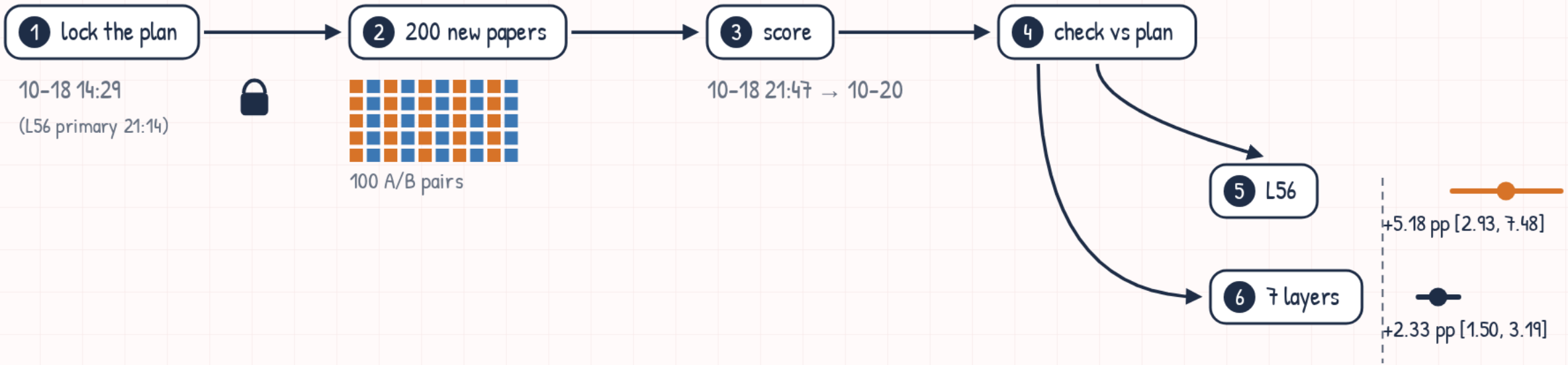
Title-adjusted A–B win rate by layer

Layer heterogeneity

L56 +0.11 but L32 **−0.03** (CI below 0).(20 pairs, matched_ab_bootstrap.csv) Reporting only the positive layers would mislead.

Independent replication (Stage 8)

plan first, data second

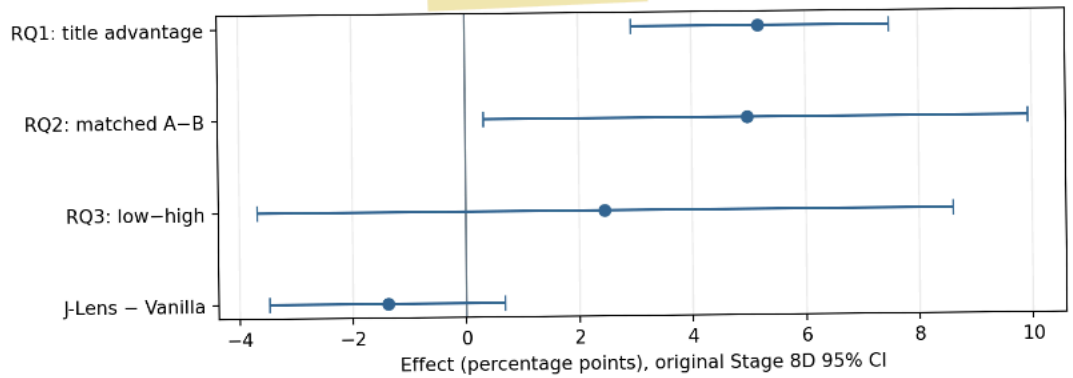


RQ1 = win rate(true title) − mean win rate(20 false titles)

first author vs 100 false names · J-Lens · paper bootstrap

+5.18 pp: Stage 8D RQ1, J-Lens L56, 200 papers, stage8D/summary/primary_questions.csv · +2.33 pp: J-Lens 7-layer, 200 papers, recompute_2026-10-30/s8_endpoints.csv

On 200 new papers the title effect replicates



Stage 8D L56 estimates (PROTOCOL.md), 95% CI (pp)(primary_questions.csv)

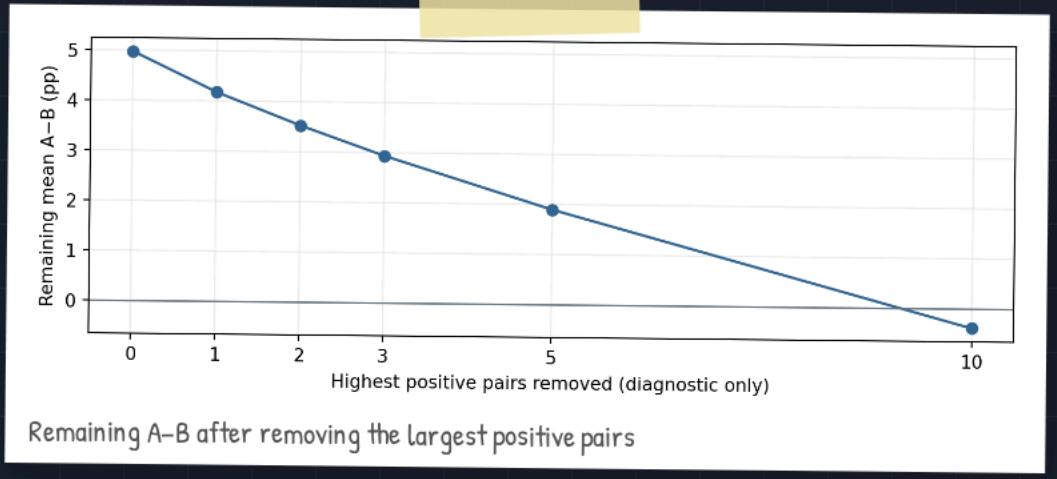
(Stage 8D, J-Lens L56; primary_questions.csv)	Estimate	95% CI
RQ1 title, L56(200 papers)	+5.18 pp	[+2.93, +7.48]
RQ2 A–B, L56(100 pairs)	+4.98 pp	[+0.32, +9.91]
RQ2 A–B, 7-layer (frozen primary)(100 pairs; s8_endpoints.csv)	+2.41 pp	[+0.91, +4.00]
RQ3 low–high(25 + 25 pairs)	+2.45 pp	[−3.69, +8.60]
J-Lens–Vanilla(jlens_minus_vanilla.csv)	−1.38 pp	[−3.48, +0.68]

Citation bands: low +6.8 [2.5, 11.6], mid +4.8, high +4.4 [−1.1, 10.0]

(50/100/50 papers; layer_and_band_summary.csv)

RQ1 is the first **independent** confirmation: correct titles help the gold author on new papers. J-Lens still adds nothing over Vanilla.

But the A–B gap rests on a few pairs



Remaining A–B after removing the largest positive pairs

- 100 pairs: 55 positive, 45 negative; median +4.4 pp (J-Lens L56; matched_pair_differences.csv)
- One pair (S8P094) ≈ +84 pp
- Remove top 5 → +1.9 pp; top 10 → −0.4 pp (recomputed from same file)

8E external audit (post-hoc)

- Unique-token weights: RQ1 +4.6, RQ2 +5.4 pp (s8_8E_weighting_by_paper.csv)
- Wrong-title length gap: +0.12 pp per token (200 papers; s8_unsourced_intervals.csv) (association only)
- Original scripts missing; recomputed from raw logits (30 Oct), same estimates (351_stage8/recompute_2026-10-30/): post-hoc only

Small samples can show a positive A–B mean that a handful of papers carry. This is the main reason Stage 10 moved to the full cohort.

Stage 8 frozen endpoint, re-scored on CPU

Addendum 13 C-2 · same 200 papers / 100 pairs · Stage 10 raw scores, all authors, all name tokens · public J-lens rank metric, NOT the W statistic

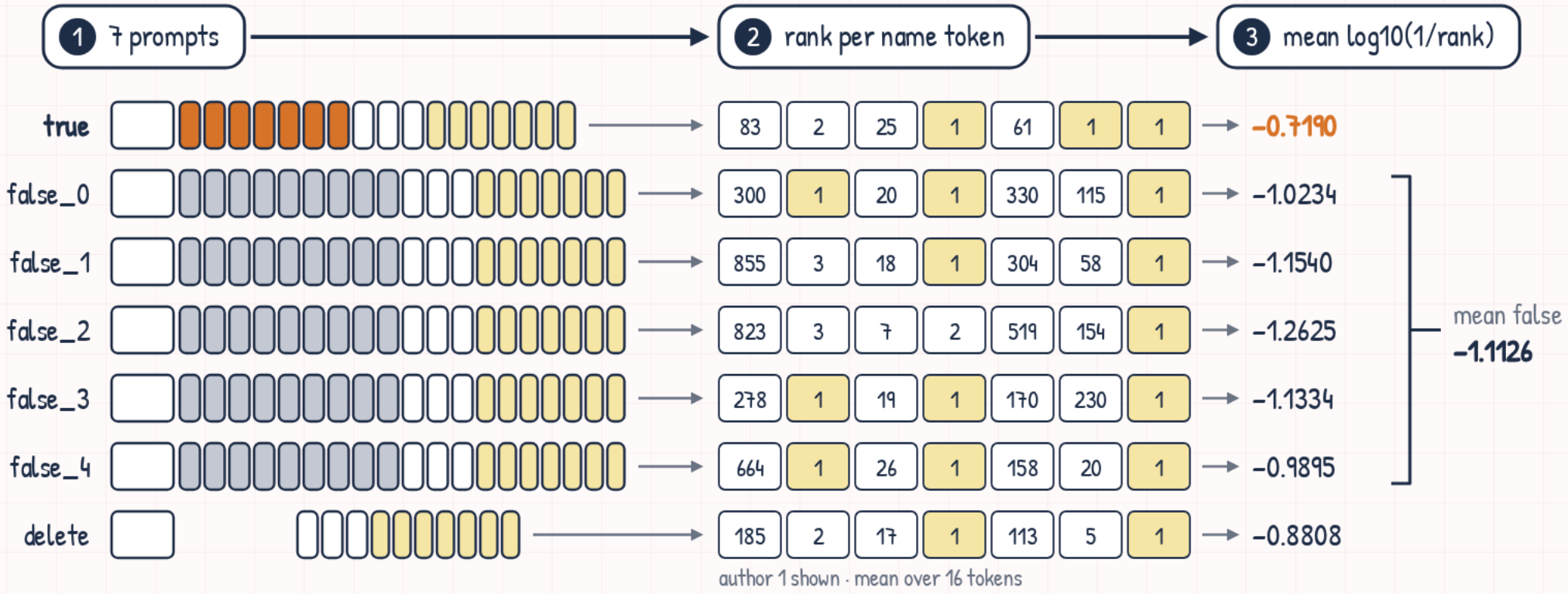
- **Multi-layer endpoint:** RQ1 +0.130 [0.108, 0.153], RQ2 +0.054 [0.008, 0.102] → both hold (public J-lens 6-layer mean; c13_C2.md aa_all)
- **L56 alone:** RQ2 +0.064 [−0.032, +0.163], not significant (same, L56)
- **What it is not:** a CPU proxy with Stage 10's 5 false titles and the rank metric; the GPU re-run of the original W was not done (user decision 2026-11-06)

Reading

The frozen multi-layer conclusion survives an all-author, all-token re-scoring; the single-layer L56 A − B gap does not.

Scoring: rank → log → spec

Coifman, Bonanno, Rafaeli (2007) · output layer



4 contrast

spec = mean nlr(true) − mean over 5 false titles

= −0.7190 − (−1.1126) = +0.3935

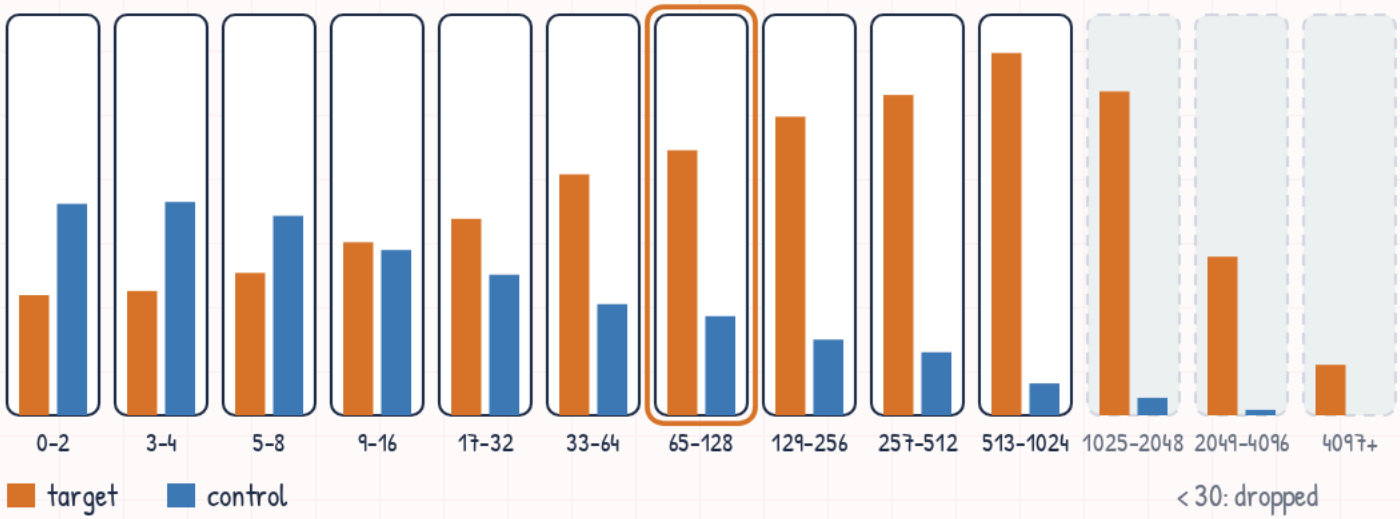
total = true − delete = +0.1618 · topic = mean false − delete = −0.2318

Ranks: 351_stage10/raw/[cond_*/]final/L63/shard_011.npz rows 25025–40 (author 1 shown) · nlr, spec, total, topic: summary_title/paper_title_contrasts.parquet (final, L63)

Statistical conventions

example paper: bin 65–128

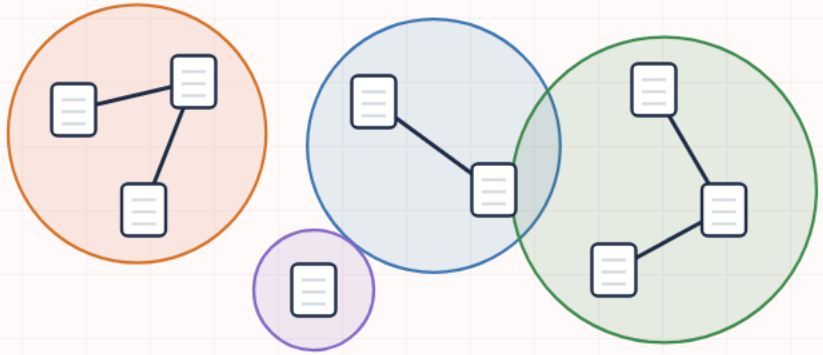
1 citation bins → 2 target − control → 3 weighted mean



< 30: dropped

$$\Delta = \Sigma\, w_b \cdot (\text{mean}_t - \text{mean}_c),\ w_b \propto n_t \cdot n_c / (n_t + n_c)$$

4 teams → 5 resample teams



shared author name = one team

7,247 teams

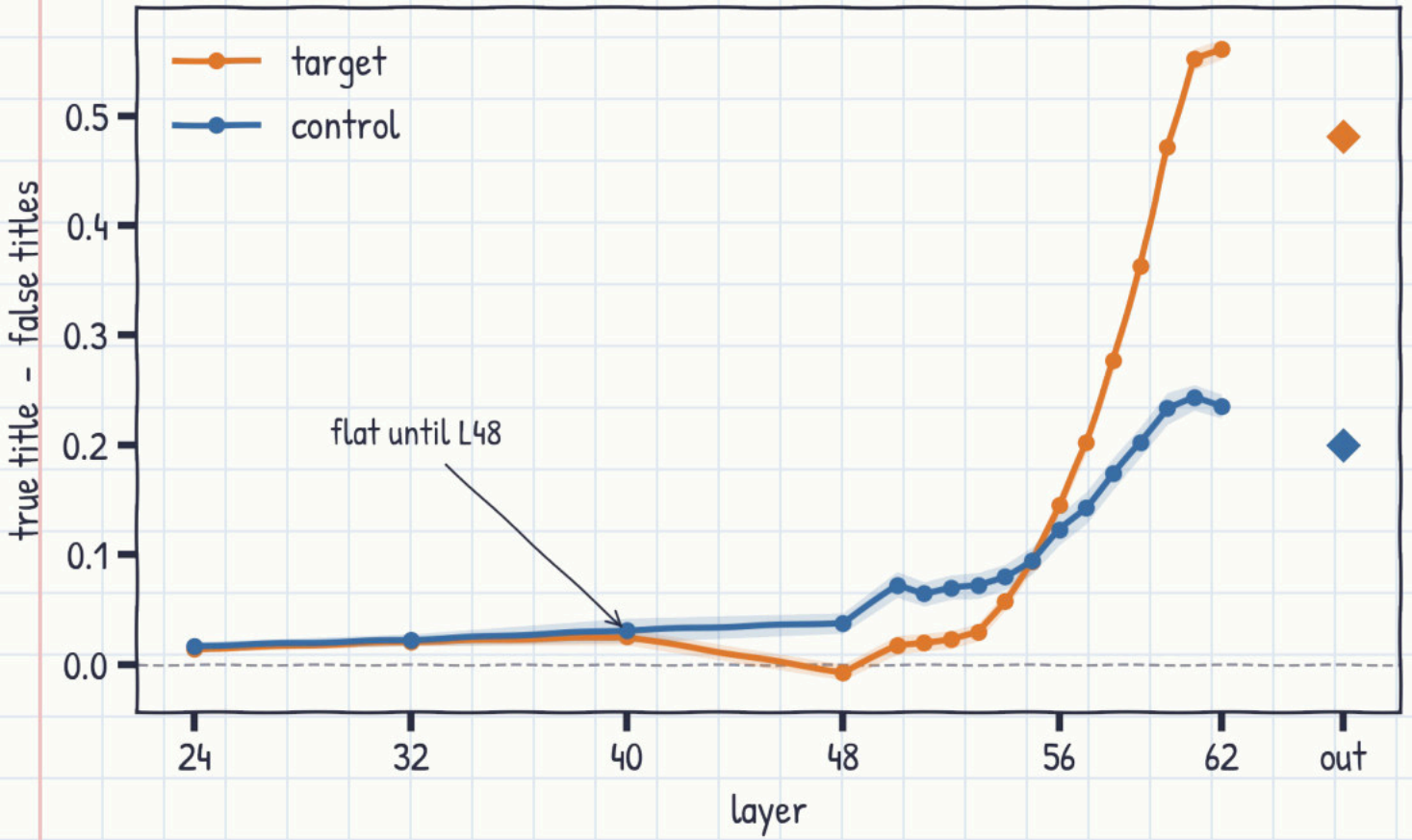


one team, one fold · 2,000 draws

Stage 10

Knowing the paper shows up only at the end

title-specific signal · public Jacobian lens (lines) and model output (diamonds)



L24–L48: ~0 for both groups (Stage 10, all readouts max 0.04, §6.3)

rises at **L56**

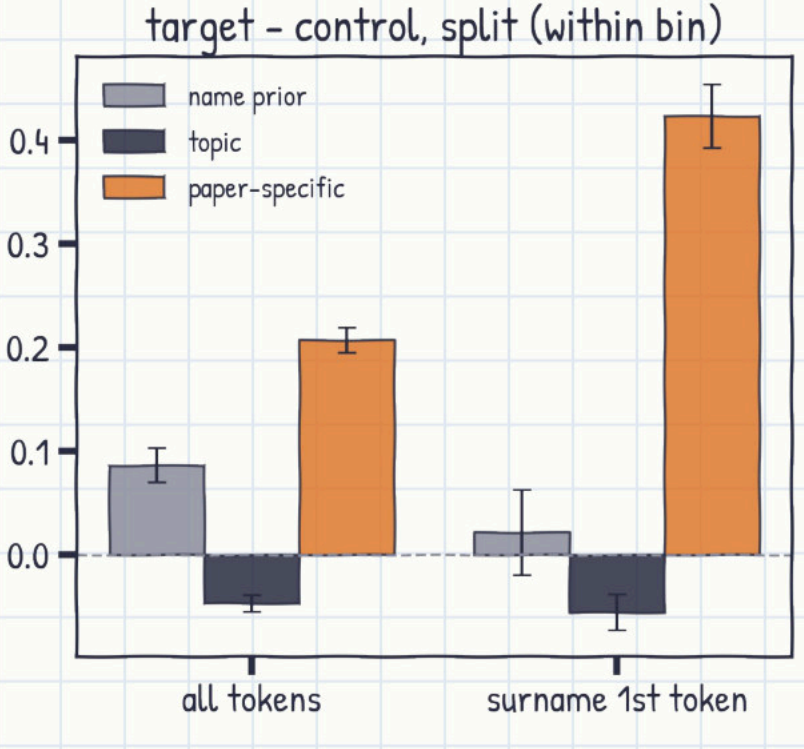
L62: **target +0.56** vs **control +0.24**
(Stage 10, public lens L62, 4,621 + 2,220 papers, §6.3 fig.)

Stage 11

How much is just the name prior?

model output · no title vs false title vs true title · all 9,108 papers





no title: rank ~12 (Stage 12 A, 9,108 papers, §6.12)

target - control: +0.09 prior, +0.21 paper (Stage 12 A, 4,621 + 2,220, within bin, §6.12)

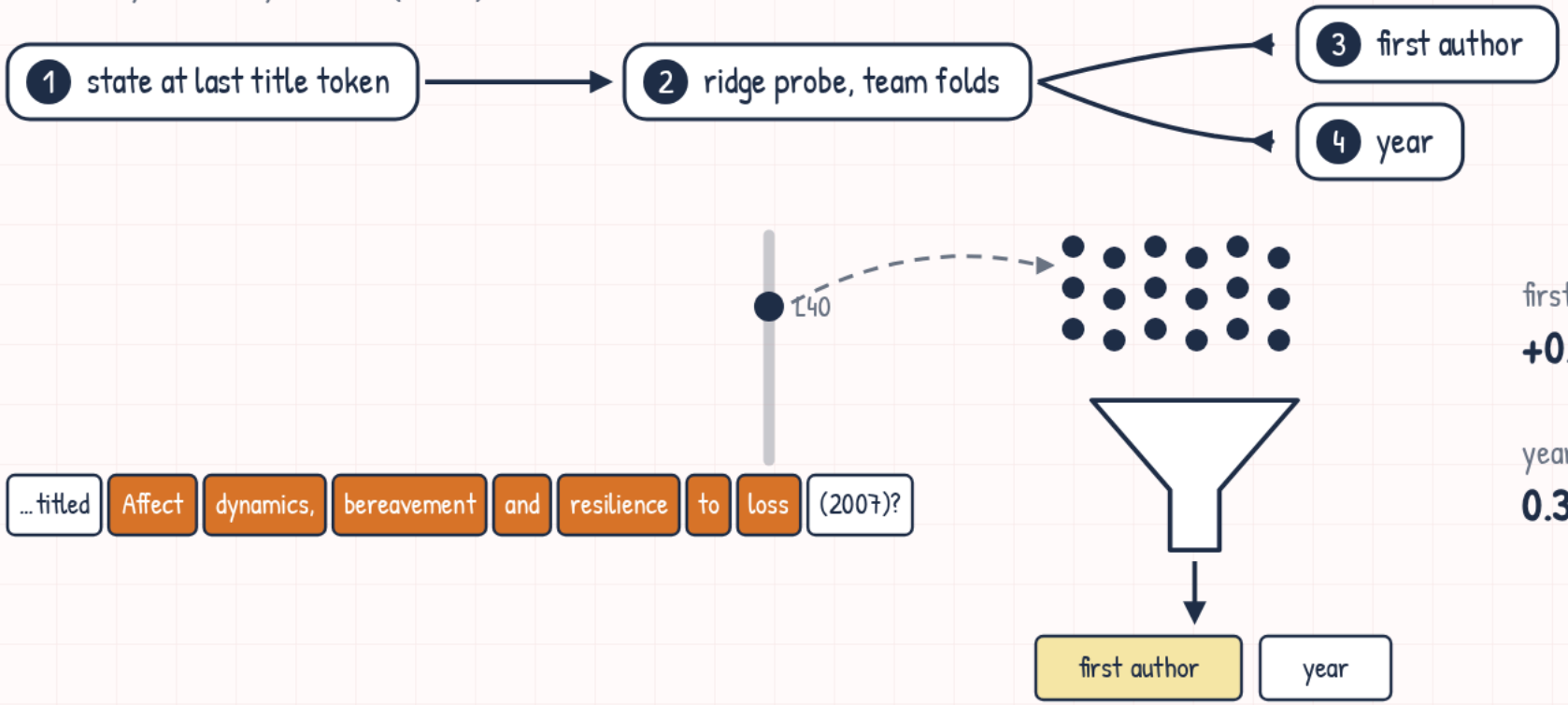
surname 1st: prior ~0, paper +0.42 (same, 1st surname token)

prior vs recognition rho 0.09 (Stage 12 B, 8,894 R0/R1 papers)

Probe

⚠ E4b scores only the first name token (mostly an initial) · all tokens by role: see the addendum-9 slide (surname first token L40 +0.071(pre-registered V0: false_0 only, 1,395 papers; e4b_alltok.md))

Coifman, Bonanno, Rafaeli (2007) · state at "loss"



first author, L40

+0.061

year R^2, L16 → L52

0.38 → 0.56



colour = author team; one team, one fold

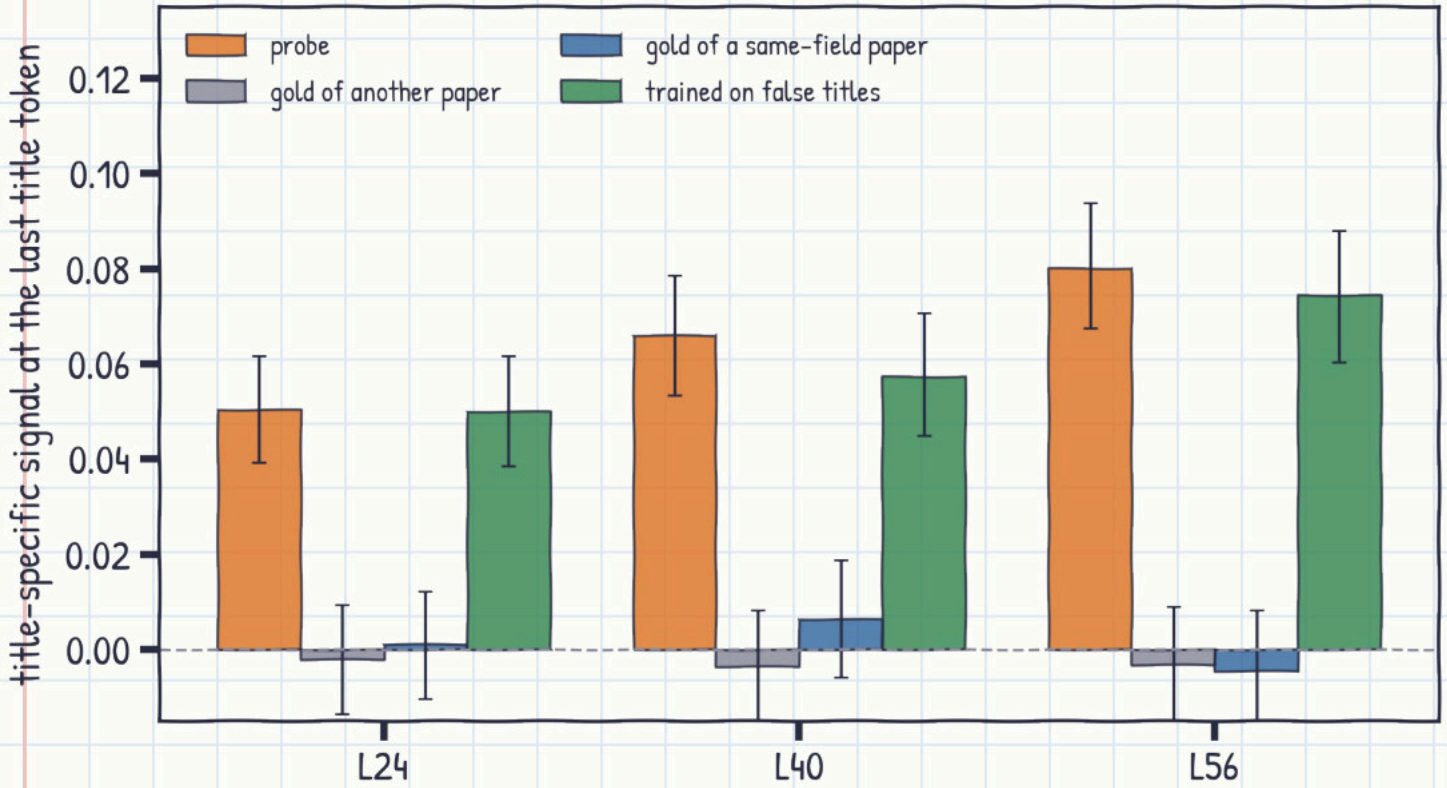
spec = probe(true title) − probe(false title)

probe score = log10(1/rank) of the true first-author token

+0.061: Stage 13 E4b re-check, probe scores ONLY THE FIRST NAME TOKEN, target + control (1,395), team folds, title reuse dropped; e4b_trainstd_compare.md
year R^2 0.38 → 0.56: Stage 13 M2, 9,108 papers, team CI; summary_cpu_teamci/M2_teamci.csv

The probe signal is paper-specific

last title token · 5-fold ridge probe · re-checked: author-team folds, title reuse removed



another paper's authors: **0**

same-field paper: **0** (Stage 12 X1 + Stage 13 re-check, §6.12)

so it is **this paper**, already at L24

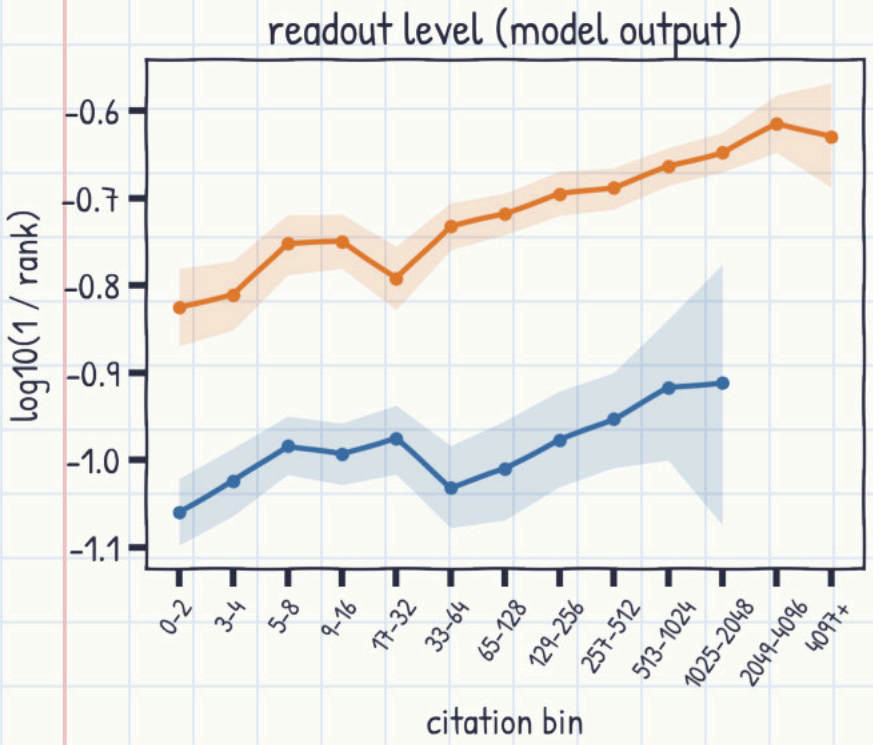
lenses cannot read it

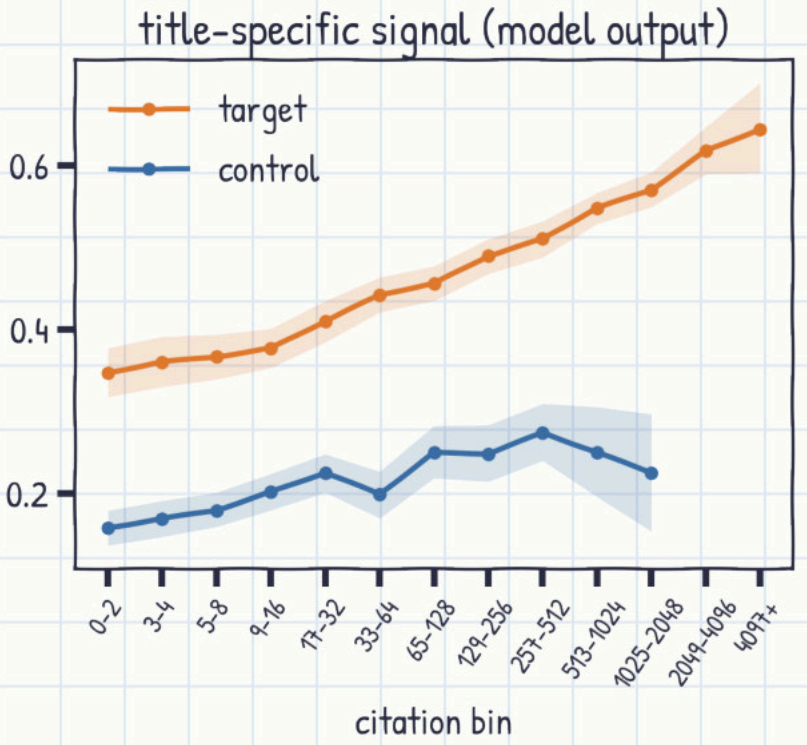
Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9, done): surname first token L40 +0.071 [0.058, 0.085] (1,395 papers; e4b_alltok.md).

Stage 10

More citations, more paper knowledge

model output · bins with < 10 papers omitted





Target: +0.35 -> +0.64 (Stage 10, final, 200 / 84 papers, §6.8)

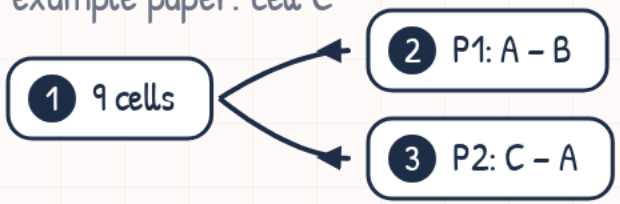
Spearman (title-specific)

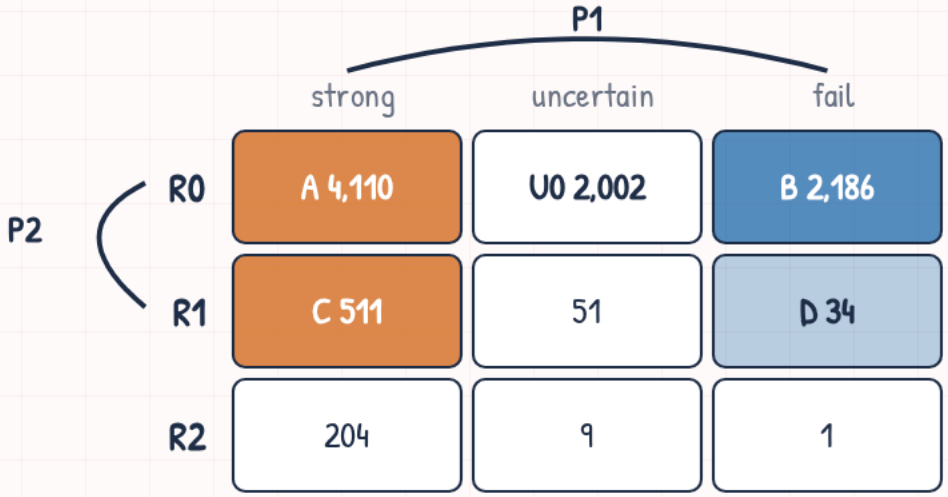
0.45 all · 0.33 target · 0.19 control (Stage 10 X2, 9,108 papers, §6.8)

Target > control in every bin

Group comparisons

example paper: cell C







P1 +0.200 [0.188, 0.213]

P2 co-authors +0.061

first author +0.031 (ns)



contrast = Σ w_b (mean_cell1 – mean_cell2)

title-specific signal, unwritten authors, output layer; team bootstrap

P1 +0.200 [0.188, 0.213]: A – B, 3,417 / 2,148 papers; P2 co-authors +0.061, first author +0.031: C – A; dose 0.17 → 0.46: R0 by orders right; 391_groups/primary.csv, stage10_curves.csv